



YOUR PHD THESIS HERE

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A thesis submitted in fulfilment of the requirements for the degree of
Doctor of Philosophy

Department of Computer Science
University College London
2025

Doctoral Candidate Declaration

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Abstract

The abstract is a very brief summary of the dissertation's contents. It should be about half a page long.

- Audience: technical or academic peers.
- Content: concise summary of the problem, methodology, results, and conclusions.
- Language: uses discipline-specific terminology and assumes the reader has technical background knowledge.
- Purpose: to give experts a quick overview of the work so they can decide whether to read the full document.

Example of Abstract

We propose a novel convex optimization framework for parameter identification in non-linear dynamical systems using sparse measurements. The approach leverages structured regularization and is validated on a benchmark biochemical reaction model, showing improved accuracy and robustness compared to existing methods.

Plain Language Summary

Also this should be about half a page long.

- Audience: non-experts, such as policymakers, journalists, or the general public.
- Content: explains why the research matters, what was done, and what the implications and impact are, without technical jargon.
- Language: clear, simple, and accessible; often includes metaphors or examples to aid understanding. Do not use acronyms.
- Purpose: to communicate the significance and relevance of the work to a broad audience.

Example of Plain Language Summary (for the same project in the abstract!)

Can you see how different this is?)

Understanding how complex systems like biological reactions behave is important in medicine and engineering. We developed a new mathematical technique that can help uncover how these systems work, even if we only have limited data. This could help researchers make better predictions and design more effective treatments.

Impact Statement

This thesis investigates [briefly state the research area and problem]. Its potential impact lies in [describe the main academic, technical, societal, or practical beneficiaries].

Beyond the immediate research contributions, the work may support [describe future systems, methods, datasets, tools, standards, clinical/practical workflows, or research communities]. The thesis also identifies limitations and responsible-use considerations that should shape future deployment and evaluation.

Acknowledgments

It is usual to thank those individuals who have provided particularly useful assistance, technical or otherwise, during your project.

This is not needed, but common.

Publications and Author Contributions

This thesis incorporates material from the following publications, accepted manuscripts, or submitted manuscripts. Each relevant chapter also includes a publication note identifying the source material and summarising the author’s contribution.

Chapter	Publication	Status / venue	DOI / link
Chapter 3	Author list. “Paper 1 title.”	Published / accepted / submitted, venue, year.	DOI URL or
Chapter 4	Author list. “Paper 2 title.”	Published / accepted / submitted, venue, year.	DOI URL or
Chapter 5	Author list. “Paper 3 title.”	Published / accepted / submitted, venue, year.	DOI URL or
Chapter 6	Author list. “Paper 4 title.”	Published / accepted / submitted, venue, year.	DOI URL or

Author contributions

For each publication, state the contributions made by the doctoral candidate and by co-authors. This should include, where applicable, conceptualisation, study design, software implementation, data collection, experiment design, analysis, figure preparation, original drafting, and manuscript revision.

Paper 1. Replace this text with the specific author contribution statement for the publication used in Chapter 3.

Paper 2. Replace this text with the specific author contribution statement for the publication used in Chapter 4.

Paper 3. Replace this text with the specific author contribution statement for the publication used in Chapter 5.

Paper 4. Replace this text with the specific author contribution statement for the publication used in Chapter 6.

Use of published material

Where text, figures, tables, or data from the publications are reused, the relevant chapter cites the original source and indicates whether permission or an open licence permits reuse. If the copyright has been transferred to a publisher, replace or redact any material that cannot be included in the public deposit version of the thesis.

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1

Introduction

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1.1	Motivation	1
1.2	Research questions	1
1.3	Thesis contributions	2
1.4	Structure of the thesis	2

1.1 Motivation

Use this chapter to introduce the problem addressed by the thesis as a whole, rather than by any single paper. The opening should explain the practical or scientific setting, why the problem matters, and what gap remains despite existing work.

1.2 Research questions

State the thesis-level research questions here. For a thesis built from publications, these questions should make clear why the four paper chapters belong in one coherent doctoral project.

1.3 Thesis contributions

Summarise the main contributions of the thesis. This section should synthesise across the publications instead of repeating each paper abstract.

1.4 Structure of the thesis

Briefly explain how the thesis is organised. Chapter 2 should review the shared research landscape and position the thesis before the publication-based chapters begin. For example, Chapter 3 may introduce the first technical contribution, Chapter 4 may extend the method or evaluation, Chapter 5 may address a complementary challenge, and Chapter 6 may provide the final study or system contribution. Chapter 7 should then connect the findings across the four publications.

2

Literature Review

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2.2 3D change detection and scene understanding	4
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2.5 Datasets, evaluation, and deployment constraints	5
2.6 Summary and thesis positioning	5

2.1 Purpose and scope

Use this chapter to establish the thesis-level research context before the paper-based chapters begin. The literature review should explain the main bodies of work that the thesis builds on, compare their assumptions and limitations, and identify the gap that motivates the four publications.

For a thesis by publication, this chapter should not simply repeat the related work sections from the papers. Instead, it should synthesise the shared intellectual background across the thesis and make clear how the later paper chapters fit within that landscape.

2.2 3D change detection and scene understanding

Summarise work on 3D reconstruction, scene representation, and change detection. Organise this section by technical approach or problem setting rather than by publication date. A useful pattern is to describe the dominant paradigm, identify the assumptions it makes about data, scene stability, or supervision, and then explain why those assumptions leave space for this thesis.

Add citations to the strongest and most recent baselines in this area. Avoid presenting the literature as a list of papers; group methods by the mechanism or limitation that matters for the thesis.

2.3 Human-in-the-loop perception and interactive correction

Discuss methods that involve user feedback, interactive refinement, active learning, correction interfaces, or human-guided perception systems. The goal is to position the thesis against work where the human is treated as a source of supervision, verification, or decision-making rather than as a passive observer.

End this section by clarifying how the thesis uses human input differently from the closest prior work. The distinction should be technical and specific, for example in terms of when feedback is collected, how it changes the system state, or how it persists across time.

2.4 Extended reality interfaces for spatial understanding

Review work on augmented, mixed, or extended reality interfaces for inspecting spatial data, supporting situated decision-making, or interacting with digital representations of physical environments.

This section should connect interface design choices to the thesis problem. Focus on how spatial context, embodied interaction, visibility of uncertainty, or in-situ feedback affects the change detection workflow.

2.5 Datasets, evaluation, and deployment constraints

Summarise the evaluation practices used in the relevant literature, including common datasets, metrics, experimental protocols, and deployment assumptions. This section should make clear which aspects of the thesis can be compared directly to prior work and which require a different evaluation framing.

Use this section to surface practical constraints such as sensor noise, scene dynamics, annotation cost, user burden, latency, and transfer across environments.

2.6 Summary and thesis positioning

Conclude the literature review by restating the gap that the thesis addresses. This section should act as a bridge into Chapter 3: explain why the first paper-based chapter is the natural first technical step after the literature review.

The final paragraph should make the chapter sequence feel deliberate: Chapter 3 introduces the first contribution, Chapters 4–6 extend or evaluate the thesis from complementary angles, and Chapter 7 later integrates the findings.

3

Paper-Based Chapter 1 Title

Contents

3.1 Chapter overview	6
3.2 Background and motivation	7
3.3 Method	7
3.4 Experiments and results	7
3.5 Discussion	7
3.6 Chapter summary and transition	7

Publication note. This chapter is based on: Author list, “Paper 1 title,” venue, year, DOI/link. Replace this note with the full citation, publication status, copyright or permission statement, and a concise author contribution statement.

3.1 Chapter overview

Do not repeat the paper abstract here. Instead, use this short overview to explain what problem this chapter addresses, how it advances the thesis-level research questions, and how it connects to the surrounding chapters.

3.2 Background and motivation

Summarise only the background needed for this chapter. Add thesis-specific context that may not have appeared in the original paper, especially where it helps the reader understand the wider doctoral project.

3.3 Method

Present the method, system, study design, or analytical approach. Rewrite the paper material into a consistent thesis voice and add details that were omitted from the publication because of page limits.

3.4 Experiments and results

Report the evaluation and findings. Reuse figures or tables only when copyright and permissions allow it, and cite the original publication wherever material is reused.

3.5 Discussion

Discuss what the results show, what they do not show, and how the chapter contributes to the thesis as a whole. This section can be broader than the paper discussion.

3.6 Chapter summary and transition

End with a short synthesis of the chapter's contribution and a transition to the next paper-based chapter.

4

Paper-Based Chapter 2 Title

Contents

4.1	Chapter overview	8
4.2	Background and motivation	9
4.3	Method	9
4.4	Experiments and results	9
4.5	Discussion	9
4.6	Chapter summary and transition	9

Publication note. This chapter is based on: Author list, “Paper 2 title,” venue, year, DOI/link. Replace this note with the full citation, publication status, copyright or permission statement, and a concise author contribution statement.

4.1 Chapter overview

Explain how this publication extends, contrasts with, or complements Chapter 3. The goal is to keep the thesis argument continuous rather than presenting four isolated papers.

4.2 Background and motivation

Add the context needed to make this chapter readable as part of the thesis.

4.3 Method

Present the chapter's technical, empirical, or design contribution.

4.4 Experiments and results

Report the evaluation and findings, adding any thesis-specific analysis that strengthens the argument.

4.5 Discussion

Explain the implications and limitations of the chapter.

4.6 Chapter summary and transition

Summarise the chapter and connect it to Chapter 5.

5

Paper-Based Chapter 3 Title

Contents

5.1	Chapter overview	10
5.2	Background and motivation	11
5.3	Method	11
5.4	Experiments and results	11
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5.6	Chapter summary and transition	11

Publication note. This chapter is based on: Author list, “Paper 3 title,” venue, year, DOI/link. Replace this note with the full citation, publication status, copyright or permission statement, and a concise author contribution statement.

5.1 Chapter overview

State the role of this paper within the thesis-level argument.

5.2 Background and motivation

Provide the specific context needed for this chapter.

5.3 Method

Describe the method, study, system, or analytical framework.

5.4 Experiments and results

Present the evidence and cite the original paper when reusing published material.

5.5 Discussion

Interpret the findings in relation to the whole thesis.

5.6 Chapter summary and transition

Summarise the contribution and connect it to Chapter 6.

6

Paper-Based Chapter 4 Title

Contents

6.1	Chapter overview	12
6.2	Background and motivation	13
6.3	Method	13
6.4	Experiments and results	13
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6.6	Chapter summary and transition	13

Publication note. This chapter is based on: Author list, “Paper 4 title,” venue, year, DOI/link. Replace this note with the full citation, publication status, copyright or permission statement, and a concise author contribution statement.

6.1 Chapter overview

Explain why this final paper-based chapter completes or extends the thesis argument.

6.2 Background and motivation

Add any context needed beyond the original publication.

6.3 Method

Describe the chapter's contribution.

6.4 Experiments and results

Present the evaluation and findings.

6.5 Discussion

Discuss the implications, limitations, and links to the earlier chapters.

6.6 Chapter summary and transition

Summarise the contribution and prepare the reader for the synthesis chapter.

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Synthesis and Discussion

Contents

7.1	Cross-chapter synthesis	14
7.2	Answering the research questions	14
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7.4	Future work	15

7.1 Cross-chapter synthesis

Use this chapter to connect the four paper-based chapters into a single doctoral argument. The synthesis should make explicit what the combined body of work establishes that no single paper establishes alone.

7.2 Answering the research questions

Return to the research questions from Chapter 1 and answer them using evidence from Chapters 3–6.

7.3 Limitations

Discuss limitations across the whole thesis. This should include limitations inherited from individual papers and broader limitations that only become clear when the work is viewed as a combined programme of research.

7.4 Future work

Identify research directions that follow from the whole thesis, not only from the final paper.

Conclusions and future directions

All good projects conclude with an objective evaluation of the project's successes and failures and suggestions for future work which can take the project further. It is important to understand that there is no such thing as a perfect project.

Even the very best pieces of work have their limitations and you are expected to provide a proper critical appraisal of what you have done. Your assessors are bound to spot the limitations of your work and you are expected to be able to do the same.



Title of the Appendix

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason.